Table 9.1 Calorie requirement based on Weight/Age

Category	Calorie requirement
Overweight	20Kcal/kg/day
Ideal weight	20Kcal/kg/day
Underweight	20Kcal/kg/day
Elderly person above 50 years	10 % less calories for each additional decade
Children 1st year	1000 calories
For girls 1-12 years	1000+ 100 calories per year of age upto 12 years
For boys 1-12 years	1000+ 125 calories per year of age upto 12 years

Carbohydrates

Carbohydrates should provide 50-60 percent of total energy. Among this, 60-70 percent should be complex carbohydrates and 30-40 percent should be simple carbohydrates. It is the type of carbohydrate that determines the glycaemic response than the actual quantity. Simple carbohydrates tend to raise the blood glucose more than complex carbohydrates. Glucose, maltose and sucrose produce large increase in blood glucose but fructose does not. Fructose can be used as a sweetener for a diabetic. Acarbose, an enzyme inhibitor present only in wheat reduces blood glucose response.



Activity : 4

Why is wheat preferred for a Diabetic than Rice? Give reasons.

Glycaemic Index

Glycaemic Index of the food is determined by measuring the area under blood glucose curve obtained by the consumption of the test food expressed as a percentage of the area obtained by giving the same quantity of carbohydrates in the form of glucose.

Fibre

High fibre diet intake improves glycaemic control and reduces insulin requirements. Patients using more fibre have exhibited reductions in fasting blood sugar levels and glycosuria and increased sensitivity to insulin. It is the soluble fibres that are most effective in producing favourable effect in carbohydrate

Table 9.2 GLYCAEMIC INDEX OF SOME COMMON FOODS

Item	Glycaemic index	Item	Glycaemic index
Bread	70	Rice	72
Wheat	70	Idli	80
Sundal	80	Milk	33
Ice cream	36	Curds	36
Groundnuts	13	Tomato soup	38
Apple	39	Banana	69
Orange	40	Beans	79
Glucose	100	Honey	87
Green gram	29	Channa	29

UNIT 9_new.indd 149 25-02-2019 15:37:14